Presentation 5

Links to Presentation and Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/SeniorMenuAnalysis_10Dec_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/NovCarDataImport_Dec10_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/SeniorMenuPresentation_Dec10_Group2Aggregate.pdf

• What did you do?

This week I worked on looking at the new data post-implementation of removing the extra senior confirmation per last year's suggestion. I analyzed any cases in which callers visited the senior menu more than once in their call journey, which would indicate to us that there is potential confusion occurring. I started by editing my previous data import code to work for the new CAR data set and imported the data. Once I had the data imported, I arranged the data by activity timestamp and contact session id and then found all cases of callers who had instances of SeniorMenu as an activity after having SuburbsOrCityMenu, meaning that they were on their second visit to the senior menu. I did the same for the old Car data. Then I found the proportion of callers who visited the senior menu multiple times of all callers who visited the senior menu and compared between the two datasets. Additionally, I calculated the median durations for these call instances.

• How does it help the project?

This gives us an idea on whether callers are getting confused on how to get to the senior menu and if they are having to reverse back through the call journey. I found that the percentage increased from 0.08% to 0.4%, which is an extremely insignificant number, but it is a factor of 5 larger than before. This indicates that there may be some extra confusion since the new implementation. However, I found that the median call time was around 8 minutes for pre and post implementation, so it doesn't seem like these calls are visiting more menus or taking much longer than before.

• Next steps

Since this is the last presentation, there are no more steps to take, but had we had more time I would suggest to try to find a way to quantify callers who ended up navigating to the wrong menu when they were intending to go to the senior menu or vice versa.

Presentation 4

Links to Presentation and Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsTerminationReasons_19Nov_Group2.pdf

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/CallTimeAnalysis_Nov19_JakeMiller.ipynb

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/SystemAnalysis_19Nov_LoganRoever.R

• What did you do?

This week we originally set out to build a predictive model that would give us a 95% prediction interval for the queue wait time to speak to an agent. I tried to build this using the Tidymodels framework in R using the cases of callers who waited in the queue to speak to an agent. Our group realized that this was giving us a large interval that wasn't useful for Legal Aid since there were so few cases of callers waiting in the queue and actually reaching an agent. We then pivoted and decided to investigate why exactly so few callers were reaching an agent when they waited in the queue and how the incoming calls were being distributed across workers. We worked together to decide what cases are considered valid call sessions and to find when calls actually have an agent's name attached. After finding all of the calls that reached an agent, we found that the mean call time was around 9 minutes for all calls.

• How does it help the project?

We thought this helped the project in 2 ways. For one, it showed that callbacks were by far the most common way that any caller was able to connect to an agent, and that waiting in the queues rarely led to any contact with the agency. Secondly, it led us to the conclusion that less time is spent connecting with callers than we expected as a whole.

• Next steps

We want to ask Cynthia about worker task distribution and how long they are expected to actually be connecting with callers every day, as it seems way too short with this system right now (given that it is tracking correctly).

Presentation 3

Links to Power BI Dashboard and Presentation:

https://app.powerbi.com/links/vMmpxeKhd6?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsCallbackPostQueue_4Nov_Group2.pdf

• What did you do?

This week, our objective was to analyze the queue average and median waiting times based on three scenarios: someone abandoned the queue, someone waited in the queue until they talked to an agent, or someone requested a callback. Jake outlined in his code that we would classify that if a contact session id had entered the queue and had "CourtesyCallback" in their flow name, that we would classify them as having requested a callback. If a caller had entered a queue, never had "CourtesyCallback" in their flow, but had "LegalServerScreenPop" as one of their activity names, then this was someone who had waited in the queue until they reached an agent. All other cases were treated as people who abandoned the queue. I looked over his code and found that it was correct and made smart assumptions based on what we know. We then created a dashboard that had the median and average wait time per queue that can be filtered by the caller type (abandoned, waited, callback) that could also be filtered by the date and time of day. On another tab, we set up a dashboard that showed the percent of caller type per queue name that could also be filtered by date and time. I worked with Jake to make the content understandable to an outside audience and to add the time component to look at trends over time.

• How does it help the project?

The trends in queue wait times and traffic based on caller type can help us understand resource allocation better and give targeted suggestions to Cynthia and Legal Aid. We also can see from the trends between 2024 and 2025 that a larger proportion of people are requesting a callback rather than waiting in the queue to talk to an agent, which tells us that Legal Aid is having success in steering people away from waiting in long queues. On the other hand, this could potentially mean they are reaching less people if the rate of success on callbacks is low.

• Next steps

Our next steps are to separate those who requested a callback and received a successful callback from those who did not receive successful callback. This will give us an idea if the increase in callback requests is helpful and how long people who do not receive a callback wait in the queue vs those who do.

Presentation 2

Links to UPDATED Power BI Dashboard and UPDATED Code:

https://app.powerbi.com/links/AsaXOFoUJO?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/Queue_Times_by_Factors_Oct28_Logan.R

Links to Power BI Dashboard, Dashboard Code, Presentation:

https://app.powerbi.com/links/KHPs-Jo0ab?ctid=cc161ac2-10c5-420b-9b80-66b398123070&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/QueueTimeSummaries_Oct21_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/QueueAnalysis_NumberAnalysis_Oct21_AggregateTeam2.pdf

• What did you do?

This week, our objective was to analyze queue waiting times per menu. To begin, I worked with Jake to define the queue names and the largest queues to analyze. Building off his work in Python, I edited the code to be usable in R which defined the calls that reached queues and were in queues with more than 500 entries (to focus resources on the largest queues). From there we calculated the average and total time in the queue for each contact session ID. To build a further analysis beyond just the counts, I joined the chosen calls back to the full dataset. I then created columns for month name, year, open/closed, and queue wait times in minutes. These will make it easier to extrapolate interpretations of the data from visualizations and filtering.

• How does it help the project?

This analysis told us how resource strain varies across months as well as across menus. While overall the most resources should be allocated to the family and housing menus, there could be rotating staff that handle the increased call volume for education, consumer, and immigration menus throughout the year. For example, education queue average wait duration increases in July with the upcoming school year, having one phone staff member focus on these calls may help reduce queue wait times. This can be applied to April with consumer queue menus and beyond.

• Next steps

My next steps for presentation 3 are to further quantify the traffic and queue wait times per menu. Looking deeply into the submenus and breaking each menu like family and housing into its submenu parts will give a better idea of which queues people are waiting the longest in. This can help with additional resource allocation and better, actionable recommendations.

Presentation 1

Links to Power BI Dashboard, Dashboard code, and Data Import Code:

• https://app.powerbi.com/links/pyRtXmrU-V?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/AggregateDataSummaries_Oct7_LoganRoever.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data import/CarDataImport_Oct7_LoganRoever.R_

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/allcallsimport_Oct7_LoganRoever.R

• What did you do?

For the first presentation, I wanted to look at which actions were happening multiple times per call. I started by converting the provided import data code to use in R instead and imported the data from there. I cleaned the data names and decided to calculate a metric which would explore the call activity distribution. For this analysis, I chose to look at calls between 10 - 75 entries because I wanted to avoid sessions that had random selections. The highest number of entries was 262, which we can deduce is from random selection to reach an agent rather than following the menu flow. I also started with 10 entries because I wanted to look at a section of calls that were inefficient, but likely valid attempts, to see where inefficiencies are occurring in the call flow. After identifying the contact session ids that matched this number of entries, I grouped by activity name. From there I calculated the total number of entries for that activity, the number of distinct sessions that selected that activity, and the average times per session the activity was selected.

• How does it help the project?

This should give us an idea of where people are getting "stuck" in an action or where the system is inefficiently directing the call to the same place again. Looking at the activities with more than 1 instance per session should help us identify possible ways to clean up the process and clear confusion / improve efficiency.

• Issues faced (if any)

We were unsure how to combine into one dashboard without the full PowerBI access, so we don't have a single dashboard yet this week. Additionally, there are some entries that I wasn't sure about, such as playmoh300s. When does this occur in a call? Is this just the hold activity?

• Attempts to resolve issues

We discussed our individual plans and got a general understanding of each other's work for this presentation. We chose one person to present all the dashboards so we wouldn't have to switch devices.

• Next steps

My next steps for presentation 2 are to analyze the QueueMenu1 activity and understand which menus have the longest queues. We will try to plot the average waiting time per queue to understand which menus need more attention and resources. The potential issues with this approach using the CAR data is that it may be difficult to identify which menu the customer was in before being sent to the queue.